Automatic Vehicle Number Plate Detection and Recognition System Using Deep Learning

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Abstract

Automatic Vehicle Number Plate Detection and Recognition is an important application of computer vision and artificial intelligence.

This project presents a deep learning based system for detecting vehicle number plates and recognizing characters printed on them.

The system uses **YOLOv8 object detection model** for locating number plates in vehicle images.

After detecting the number plate region, **EasyOCR** is applied to extract text from the cropped plate image.

The system can be used in traffic monitoring, security surveillance, parking management, and electronic challan systems.

The main advantage of this system is automation of vehicle identification without human intervention.

Introduction

In modern society, traffic management and vehicle monitoring have become important requirements due to the increasing number of vehicles on roads. Manual vehicle identification is time consuming and prone to human errors. Therefore, automatic recognition systems are developed using computer vision and artificial intelligence techniques.

Automatic Vehicle Number Plate Detection and Recognition System is an intelligent application that automatically detects vehicle registration plates from images and extracts characters printed on them.

This project is implemented using deep learning technology, specifically the **YOLOv8 object detection model**. YOLO stands for *You Only Look Once*, which is a real-time object detection algorithm capable of detecting objects quickly and accurately.

After detecting the number plate region, Optical Character Recognition (OCR) technology is used to convert image characters into machine readable text. In this project, **EasyOCR** is used for text recognition.

The system is useful in traffic law enforcement, security monitoring, parking management, and smart city surveillance systems. The development of such automated systems helps reduce human workload and increases accuracy in vehicle identification.

Problem Statement

Vehicle identification is an important task in modern transportation systems. Traditional methods of checking vehicle registration numbers are manual and inefficient.

The major problems associated with manual vehicle number identification include time consumption, human errors, and difficulty in monitoring large traffic areas.

In busy traffic environments, it is difficult for human operators to continuously monitor vehicles and record registration numbers accurately.

There is a need for an automated system that can detect vehicle number plates and extract registration information without human intervention.

This project aims to solve these problems by implementing an automatic vehicle number plate detection and recognition system using deep learning and OCR technology.

Objectives of the Project

The main objectives of this project are:

- To develop an automatic vehicle number plate detection system.
- To train a deep learning model using a labeled dataset.
- To detect vehicle number plates from vehicle images.
- To extract characters from detected number plates using OCR technology.
- To display detected number plate information on the image.
- To demonstrate practical implementation of computer vision technology.

The system is designed to provide an automated solution for vehicle identification.

Tools and Technologies Used

The following tools and technologies were used in the development of this project.

Programming Language

Python was used as the primary programming language for implementation.

Deep Learning Framework

YOLOv8 (You Only Look Once version 8) was used for object detection.

Image Processing Library

OpenCV library was used for image processing operations such as cropping and drawing bounding boxes.

Optical Character Recognition

EasyOCR was used to extract text from detected number plates.

Visualization Tool

Matplotlib library was used to display detection results.

Development Environment

Google Colab was used for training and testing the model.

GitHub Version Control

GitHub was used for version control and backup of project code.

System Architecture

The system architecture of the Automatic Vehicle Number Plate Detection and Recognition System consists of several processing stages.

The system follows a pipeline approach where input data passes through multiple modules for processing.

Input Image Acquisition

The system accepts vehicle images as input.

Number Plate Detection using YOLOv8

The trained YOLOv8 model detects the license plate region from the vehicle image.

Plate Cropping

After detection, the number plate area is cropped from the original image.

Optical Character Recognition (OCR)

EasyOCR is applied to recognize characters from the cropped plate image.

Output Display

The detected number plate and recognized text are displayed on the screen.

This architecture ensures efficient and accurate vehicle identification.

System Architecture Diagram

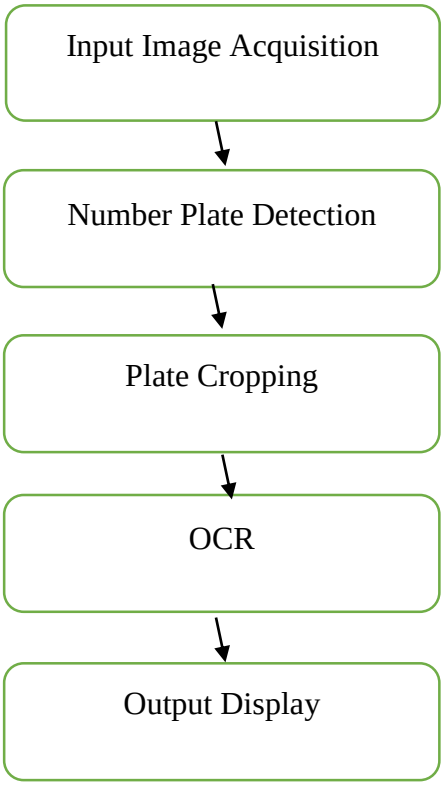

Figure 1

System Modules Description

The project consists of the following main modules:

Dataset Preparation Module

This module contains vehicle images used for training and testing the model.

The dataset is stored in YOLO annotation format.

Model Training Module

In this module, the YOLOv8 deep learning model is trained using labeled dataset images.

The model learns to detect number plate regions from vehicle images.

Detection Module

This module is responsible for detecting number plates from input images.

Bounding boxes are drawn around detected number plates.

Text Recognition Module

After detection, OCR technology is used to extract characters from the plate region.

EasyOCR library is used for character recognition.

Output Display Module

The final output is displayed by showing detected number plate and recognized text.

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Figure 2

Implementation Details

The Automatic Vehicle Number Plate Detection and Recognition System was implemented using Python programming language.

The project uses deep learning and computer vision techniques for vehicle plate detection.

Dataset Usage

The project dataset was stored in YOLO format containing training and testing images.

The dataset was used to train the YOLOv8 object detection model.

Model Training

The pretrained YOLOv8 model was fine-tuned using the license plate dataset.

Training was performed for 50 epochs with image size 640.

After training, the best model weights were saved as **best.pt**.

Detection Process

The implementation process includes the following steps:

- Load trained YOLO model.
- Input vehicle image.
- Detect number plate region.
- Crop plate image.
- Apply OCR for text extraction.
- Display output.

Figure 3

Advantages of the System

The Automatic Vehicle Number Plate Detection and Recognition System provides several advantages:

- The system is fully automatic and reduces human effort.
- It improves vehicle identification speed.
- It can be used in traffic surveillance systems.
- It reduces manual monitoring errors.
- It is cost effective compared to traditional monitoring methods.
- The system can work on real vehicle images.
- It is useful for smart city traffic management.
- It can be integrated with security and e-challan systems.

Limitations of the System

Although the system provides automated detection, it also has some limitations.

- The accuracy of OCR depends on image quality.
- Blurred or low-resolution images may reduce recognition performance.
- Plates with complex backgrounds are difficult to detect.
- Very small or tilted number plates may cause detection errors.
- The system is trained on a limited dataset.
- Real-time video processing was not fully implemented.
- Performance may vary under poor lighting conditions.

Testing and Results

The developed system was tested using vehicle images from the dataset and external test images.

The YOLOv8 trained model was used to detect number plates from input car images.

After detection, the plate region was cropped and Optical Character Recognition (OCR) was applied to extract text.

The system successfully detected number plates in most clear images.

The detection performance depends on image quality, lighting conditions, and plate visibility.

Result Observation

- Number plate regions were detected using bounding boxes.
- OCR successfully extracted characters from detected plates.
- Detected text was displayed on the image output.

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Figures: Testing Screens

Conclusion

The Automatic Vehicle Number Plate Detection and Recognition System was successfully developed using deep learning and computer vision techniques.

The system uses YOLOv8 object detection model for locating vehicle number plates and EasyOCR for character recognition.

The project demonstrates practical implementation of artificial intelligence in traffic monitoring and vehicle identification.

The system can automatically detect number plates from vehicle images and extract registration information.

Although the system accuracy depends on image quality, the project provides a strong foundation for automated vehicle recognition systems.

Future improvements can be made by using larger datasets and real-time video processing.

References

The following resources were used during the development of the Automatic Vehicle Number Plate Detection and Recognition System Using Deep Learning:

- Ultralytics YOLOv8 Documentation
  https://docs.ultralytics.com
- OpenCV Official Documentation
  https://opencv.org
- EasyOCR GitHub Repository
  https://github.com/JaidedAI/EasyOCR
- Kaggle Dataset Resources
  https://www.kaggle.com
- Research papers on Automatic Number Plate Recognition systems.
- Python Official Documentation
  https://www.python.org

Appendix

The appendix section contains additional project-related information.

Project Source Code

The project was developed using Python programming language.

Major libraries used in the project include:

OpenCV

YOLOv8 (Ultralytics framework)

EasyOCR

Matplotlib

Dataset Information

The dataset used in this project is **License-Plate-Data** which contains:

- Training images
  Testing images
  Annotation labels in YOLO format

Training Details

- Model Used: YOLOv8n
  Epochs: 50
  Image Size: 640
  Detection Class: License Plate